FULL NAME: ______________________ **BLOCK:** __________

Radical Equations With Rational Exponents + DOK-3 Modeling

Lesson 5-4 · Period 2 · Student Packet

OBJECTIVES

Math Objective	Solve equations with rational exponents by converting to radical form and raising both sides to the reciprocal power; apply rational-exponent models to real-world half-life and decay problems.
Language Objective	Translate a real-world half-life scenario into an equation with a rational exponent, compute, and interpret the result in context.
Essential Understanding	Rational exponents are the bridge between algebraic power operations and real-world exponential/radical modeling — the half-life formula $y = a \cdot (1/2)^{(t/h)}$ uses fractional powers directly.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Solve rational-exponent equations; model real-world decay with fractional-power formulas.
Essential Question	How do rational-exponent equations model real-world decay, and when do extraneous solutions arise?
Materials	Student packet, pencil, calculator for Q#41. Teacher models Ex 4; DOK-3 Practice #41 is the topic's capstone modeling task.

LAUNCH — Rational-Exponent Equations (teacher models Example 4)

[BOOK] RATIONAL-EXPONENT RULES (keep printed on your page)

RATIONAL-EXPONENT RULES

1. Rewrite $x^{(m/n)}$ as $(\sqrt[n]{x})^m$ — bridge to radicals you already know.
2. To solve $(\text{expr})^{(m/n)} = k$, raise both sides to the reciprocal power n/m.
3. Simplify, then CHECK — even rational-exponent equations can have extraneous solutions when the denominator of the exponent is EVEN.
4. For real-world decay $y = a \cdot (0.5)^{(t/h)}$: half-life h tells you the exponent's denominator; plug in t, evaluate.

[WARN] EXTRANEOUS SOLUTIONS ARISE WHEN THE DENOMINATOR IS EVEN

$(\text{expr})^{(m/n)} = k \rightarrow$ raise both sides to n/m
If n is EVEN, squaring can introduce solutions that don't satisfy the original equation.
ALWAYS substitute back into the ORIGINAL equation to verify.

EXPLORE — Think-Pair-Share (33 min)

Work with a partner. Move in order. Practice #41 (Half-Life Model) is the big one — plan ~12 min for it.

Bank-item labels (in order):

1. Try It 4 — Rational-exponent equation
2. Practice #12 — Solve with rational exponent
3. Practice #33 — Rational-exponent equation
4. Practice #34 — Rational-exponent equation
5. Practice #41 [STAR] DOK-3 HALF-LIFE MODEL

Try It 4 — Rational-exponent equation

Solve each equation.

a. $(x^2 - 3x - 6)^{3/2} - 14 = -6$

b. $(x - 8)^{3/2} = (10 - x)^3$

Practice #12 — Solve with rational exponent

$(3x + 2)^{2/3} = 4$

Practice #33 — Rational-exponent equation

Solve. $(0.5(x^2 + 5x + 136)^{(2)/(3)} = 50)$

Practice #34 — Rational-exponent equation

Solve. $(2(x^2 - 12x - 4)^{(1)/(2)} - 3 = 15)$

Practice #41 [STAR] DOK-3 HALF-LIFE MODEL

Make Sense and Persevere The half-life of a certain type of soft drink is 5 h. If you drink 50 mL of this drink, the formula $y = 50(0.5)^{(t)/(5)}$ tells the amount of the drink left in your system after t hours. How much of the soft drink will be left in your system after 16 hours?

SHARE / SUMMARY (5 min)

Sentence frames

Pair share (Half-Life Model): "This equation has exponent ___/___, so I raise both sides to the power ___/___. After simplifying I get ___. In the half-life formula, $a =$ ___ (start), $h =$ ___ (half-life period), $t =$ ___ (elapsed time), giving $y \approx$ ___."

Whole class: one pair shares their computation and interpretation. Class signals thumbs up/sideways.

[SEARCH] BRIDGE TO P3

Next period we apply rational-exponent skills to assessment-critical items. Bring your check-for-extraneous habit.

EXIT

To solve $(\text{expr})^{(m/n)} = k$ I raise both sides to power ___.
Half-life tells me the ___ of the exponent.
After 16 hours, 50 mL of soft drink decays to ___ mL.
One biggest thing I learned today: ____________________________

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #19 is a conceptual extraneous-solution argument that extends the half-life skill.

Optional #1 (Savvas Practice, extraneous-solution argument)

Higher Order Thinking Some, but not all, equations with rational exponents have extraneous solutions. What is the relationship between the exponents and the possibility of having extraneous solutions for equations with rational exponents? Explain your reasoning.